

AI-Based Real-Time Helmet Violation Detection and Automatic E-Challan Generation Using YOLO, OCR, and Deep Learning

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: AI-Based Real-Time Helmet Violation Detection and Automatic E-Challan Generation Using YOLO, OCR, and Deep Learning

1. Abstract:

Road accidents caused by riders not wearing helmets remain a serious concern. Manual monitoring of traffic violations is time-consuming and often inaccurate. This project proposes an AI-Based Real-Time Helmet Violation Detection and Automatic E-Challan Generation System using deep learning and computer vision. The system uses the YOLO object detection model to identify motorcycles and detect whether riders are wearing helmets from CCTV footage or recorded videos. If a helmet violation is detected, the vehicle's license plate is extracted and recognized using Optical Character Recognition (OCR). The recognized vehicle number, along with the captured image, date, and time, is stored in a database, and an automatic e-challan is generated. The system also provides an administrator dashboard to monitor and manage detected violations. By automating the entire process, the proposed solution reduces manual effort, improves detection accuracy, and enables faster enforcement of traffic rules. This project contributes to smarter traffic management systems and encourages helmet usage, ultimately helping improve road safety.

2. INTRODUCTION:

Road accidents caused by riders not wearing helmets remain a major safety concern and contribute significantly to serious injuries and loss of life. Despite traffic rules and awareness campaigns, many riders continue to violate helmet regulations. Manual monitoring of traffic violations through CCTV cameras requires significant human effort and is time-consuming, especially in busy traffic areas. It can also be prone to errors and delays in identifying violators. Therefore, there is a need for an automated and reliable system that can continuously monitor traffic and detect helmet violations in real time. The integration of Artificial Intelligence, computer vision, and deep learning provides an effective approach to addressing this problem.

This project proposes an AI-based Real-Time Helmet Violation Detection and Automatic E-Challan Generation System using YOLO object detection and Optical Character Recognition (OCR). The system analyzes CCTV footage or recorded traffic videos to identify motorcycles and determine whether riders are wearing helmets. When a rider without a helmet is detected, the system captures the vehicle and extracts its number plate. OCR technology is then used to recognize the vehicle registration number automatically. The detected violation details, including the vehicle number, captured image, date, and time, can be stored in a database and used to generate an automatic e-challan. The proposed system reduces manual intervention, improves the speed and accuracy of traffic violation detection, enables faster enforcement of traffic rules, and contributes to safer and smarter traffic management.

3.SOFTWARE REQUIREMENTS:

Software / Framework	Purpose
Python 3.11+	Programming Language
Visual Studio Code	Development Environment
YOLOv11 (Ultralytics)	Helmet & Vehicle Detection
OpenCV	Image & Video Processing
EasyOCR	Number Plate Recognition (OCR)
FastAPI	Backend API Development
MySQL	Database Management
PyTorchPyTorch	Deep Learning Framework
Uvicorn	ASGI Server for FastAPI
Git & GitHub	Version Control

4.HARDWARE REQUIREMENTS:

Hardware	Purpose
Operating System	Windows 10/11 (64-bit)
Camera	CCTV Camera / USB Webcam for Real-Time Detection
GPU	NVIDIA GPU with CUDA support for Deep Learning Model Training

Signature of the Guide

Course Instructor